



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

20-05-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	B	C	B		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	A,C	A,B,D	A,B,C,D	A,D	B	A,C,D
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.50	60.00	24.00	2.00	9.00	1900.00
SECTION-II (ii)	Q.	7	8	9			
	A.	46	20	7			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	A	C	D		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C,D	A,D	A,C,D	A,B	A,B,C,D	A,C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	5.52	9.35	9.00	11.00	0.25	0.33
SECTION-II (ii)	Q.	7	8	9			
	A.	56	1	5			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4		
	A.	B	C	A	A		
SECTION-I (ii)	Q.	5	6	7	8	9	10
	A.	B,C	A,D	B,C,D	C,D	B,D	C
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	2.71	0.25	300.00	1200.00	0.33	0.75
SECTION-II (ii)	Q.	7	8	9			
	A.	8	511	4			

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (B)

Applying Ampere's circuital law,

$$B \cdot 2\pi r = \frac{\mu_0 I \pi r^2}{\pi a^2} \text{ for } r < a$$

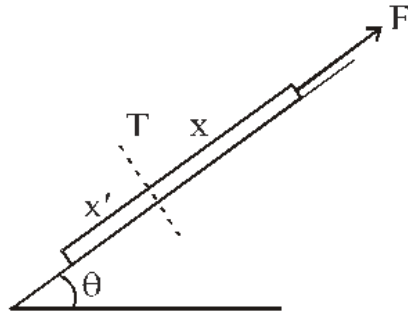
$$\text{and } B \cdot 2\pi r = \mu_0 I \text{ for } R > a$$

Taking the ratio

$$\frac{r}{R} = \frac{r^2}{a^2}$$

$$\therefore a = \sqrt{Rr}$$

2. Ans (B)



$$F - Mg \sin \theta = ma$$

$$T - \frac{Mx'}{L} g \sin \theta = \frac{Mx'}{L} a$$

$$T - \frac{Mx'}{L} g \sin \theta = (F - Mg \sin \theta) \frac{x'}{L}$$

$$T = \frac{Fx'}{L} = \frac{F(L-x)}{L}$$

PART-1 : PHYSICS

SECTION-I (ii)

8. Ans (A,D)

$$\tau = I\alpha$$

$$mgR - NR = 4mR^2\alpha \quad \dots\dots\dots(i)$$

$$mg - N = 4m\alpha R \quad \dots\dots\dots(ii)$$

$$N = m\alpha R$$

From (i) and (ii)

$$mg = 5m\alpha R$$

$$\alpha = g/5R$$

$\therefore$ acceleration of bead 'A' is

$$a_A = \alpha R \sqrt{2} = \frac{g\sqrt{2}}{5} = 2\sqrt{2}m/s^2$$

$\therefore$ acceleration of bead 'B' is

$$a_B = \frac{g}{5} \sqrt{26} = 2\sqrt{26}m/s^2$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (1.50)

Experiment - I

$$\frac{1}{72} + \frac{1}{36} = \frac{1}{f} \Rightarrow f = 24cm$$

Experiment - II

$$\frac{1}{-30} + \frac{1}{60} = -p_{eq} = \frac{-1}{20}$$

$$p_{eq} = 2p_L + p_m$$

$$\Rightarrow \frac{1}{20} = \frac{2}{24} - \frac{2}{R_1} \Rightarrow R_1 = 60cm_{convex}$$

Experiment - III

$$\frac{3}{-20} - \frac{1}{7.5} = -p_{eq} = \frac{-17}{60}$$

$$p_{eq} = 2p_L + p_m$$

$$\Rightarrow \frac{17}{60} = \frac{2}{24} - \frac{2}{R_2} \Rightarrow R_2 = -10cm$$

$$\frac{1}{24} = (\mu - 1) \left(\frac{1}{10} - \frac{1}{60} \right)$$

$$\Rightarrow \mu = 1.5$$

3. Ans (24.00)

In series, $V \propto \frac{1}{C}$

$$\Rightarrow V_2 = 8 \text{ volt}$$

$$q_2 = 8 \times 3 = 24\mu C$$

4. Ans (2.00)

$$t = \frac{T}{4} = \frac{2\pi\sqrt{LC}}{4}$$

6. Ans (1900.00)

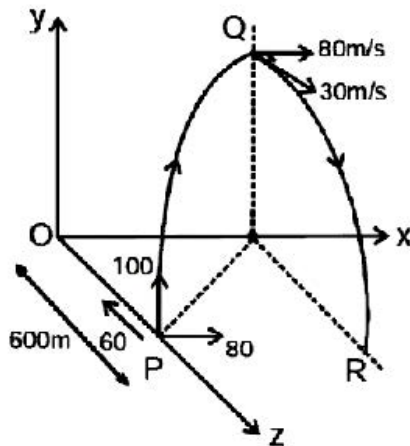
Since wall is smooth time of flight remains

unchanged

$$T = \frac{2u_y}{g} = \frac{2 \times 100}{10} = 20s$$

The particle reaches the wall

$$\text{at } t = \frac{600}{60}s = 10s$$



At $t = 10s$, the particle is at its maximum

height. Hence, $t = 10s$ and $\theta = 90^\circ$

Let Q be the point of impact on the wall.

So its X - co-ordinate = $80 \times 10 = 800m$

$$Y - \text{co-ordinate} = \frac{u_y^2}{2g} = \frac{(100)^2}{2 \times 10} = 500m$$

Z - co-ordinate = 0

$$\vec{V}_Q = 80\hat{i} + \mathbf{e} \times 60\hat{k} = (80\hat{i} + 30\hat{k}) \text{ m/s}$$

And, time to reach the XZ- plane is 10s from Q

X - co-ordinate = $80 \times 20 = 1600m$

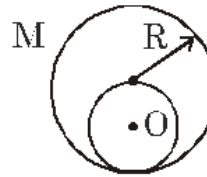
Y - co-ordinate = 0

Z - co-ordinate = $30 \times 10 = 300m$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (46)



$$I_0 = \frac{4M}{3} \left(\frac{R^2}{2} + \left(\frac{R}{2} \right)^2 \right) - \frac{M}{3 \times 2} \left(\frac{R}{2} \right)^2 = \frac{23}{24} MR^2$$

$$KE = \frac{1}{2} \left(\frac{23}{24} \right) MR^2 \omega^2$$

$$= \frac{23}{24 \times 2} \times (12)(2)^2 (\sqrt{200})^2 = 46 \text{ Joule}$$

8. Ans (20)

Let mass of the balloon be M and its volume on

the surface be v_0 .

$$Mg = \frac{v_0}{3} \rho g \quad \dots (i) \quad [\rho = \text{density of water}]$$

At depth 'h', the volume of the balloon will

decrease due to hydrostatic pressure. Let new

volume be v. $p_2 v_2 = p_1 v_1$ gives

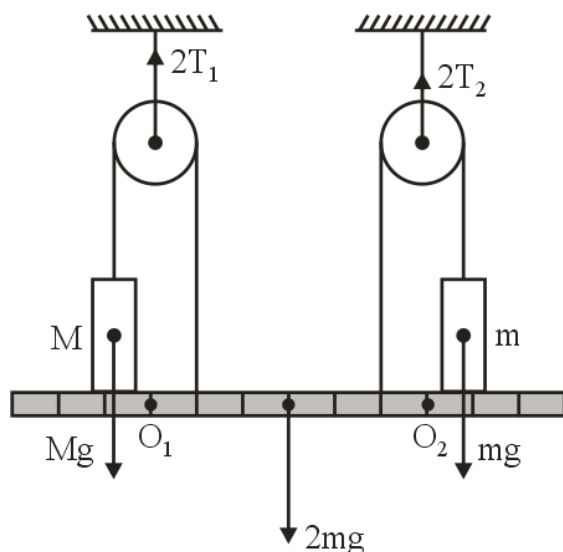
$$(h + 10)v = 10v_0 \Rightarrow v = \frac{10v_0}{h + 10}$$

for equilibrium $v\rho g = Mg$

$$\frac{10v_0}{h+10} \rho g = \frac{v_0}{3} \rho g \quad [u \sin g (i)]$$

$$\Rightarrow 30 = h + 10 \Rightarrow h = 20m$$

9. Ans (7)



$$Mgx + 2T_2 (6x) = 2mg (3x) + mg (7x)$$

$$mgx + 2T_1 (6x) = 2mg (3x) + Mg (7x)$$

$$T_1 = \frac{5m + 7M}{12}g, \quad T_2 = \frac{13m - M}{12}g$$

$$\therefore M < 13m \quad M_{\max} = 13m$$

$$T_1 + N_1 = Mg, \quad mg = T_2 + N_2$$

$$N_1 = \frac{5(M - m)g}{12}, \quad N_2 = \frac{(M - m)g}{12}$$

$$\therefore M > m \quad M_{\min} = m$$

PART-2 : CHEMISTRY

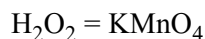
SECTION-I (i)

1. Ans (B)

$$M_{H_2O_2} = \frac{20}{11.35}$$

$$a_0 = \frac{20}{11.35} \times 11.35 = 20 \text{ millimole}$$

After 6 hour

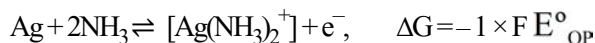
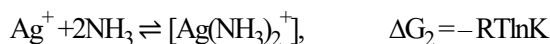


$$n_{H_2O_2} \times 2 = 0.025 \times 25 \times 5 \times 10$$

$$n_{H_2O_2} = a_t = \frac{125}{8} \text{ millimole}$$

$$K = \frac{1}{6} \ln \left(\frac{20}{125/8} \right) = 0.05 \text{ hr}^{-1}$$

2. Ans (A)



$$\Delta G = \Delta G_1 + \Delta G_2 = E^\circ_{OP} = -0.8 + \frac{2.303RT}{F} \log 10^{16} = 0.144 \text{ V}$$

3. Ans (C)

As, this is a high spin complex,

so, $\Delta_0 < P.E.$

$$CFSE = -0.4n_1 \Delta_0 + 0.6n_2 \Delta_0$$

$$7200 = (-0.4 \times 5\Delta_0) + (0.6 \times 2\Delta_0)$$

$$= -0.8\Delta_0$$

$$\Delta_0 = -9000 \text{ cm}^{-1} \quad \left| \begin{array}{l} P.E. > 9000 \text{ cm}^{-1} \\ = 9000 \text{ cm}^{-1} \end{array} \right.$$

while, the hexa complexes of Co^{3+} are generally, low spin complexes,

Hence, more Δ_0 & less λ absorbed.

4. Ans (D)

More the extent of conjugation, lesser is the heat of hydrogenation.

PART-2 : CHEMISTRY

SECTION-I (ii)

5. Ans (B,C,D)

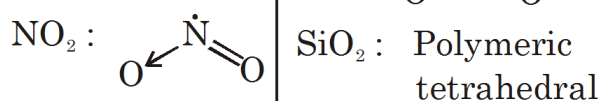
SRP $\downarrow$ spontaneity of reaction $\uparrow$

So Ag and Au remains in anode mud. When anode dissolve then first $Fe \rightarrow Fe^{+2}$ which change the blue colour intensity.

Cathode product = Cu

Anode product = Fe^{+2} , Cu^{+2} .

6. Ans (A,D)



* CO_2 , NO_2 , SO_2 are gases at room temperature.

* All except SiO_2 are water soluble.

* All are acidic oxides.

8. **Ans (A,B)**

Electron withdrawing groups tend to stabilize the carbanion formed.

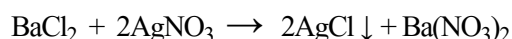
9. **Ans (A,B,C,D)**

1,2-addition & 1,4-addition rules based on nature of reagent and substrate.

10. **Ans (A,C)**

$$\Delta T_{b,A} = K_b \cdot m = 0.5 \times (0.1 \times 3) = 0.15^\circ\text{C}$$

$$\therefore \text{B.P. of solution A} = 100.15^\circ\text{C}$$



$$0.1 \quad 0.2$$

$$0 \quad 0 \quad 0.1$$

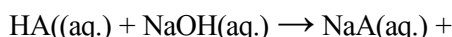
$$\text{Now, } \Delta T_{b,B} = K_b \cdot m = 0.5 \times \left(\frac{0.1}{2} \times 3 \right) = 0.075^\circ\text{C}$$

$$\therefore \Delta T_{b,A} - \Delta T_{b,B} = 0.15 - 0.075 = 0.075^\circ\text{C}$$

PART-2 : CHEMISTRY

SECTION-II (i)

1. **Ans (5.52)**



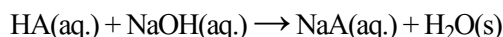
$$\text{m moles} \quad 10 \quad 2.5$$

$$\text{m moles} \quad 7.5 \quad - \quad 2.5 \quad 2.5$$

$\therefore$ Resulting solution HA(aq.) and NaA(aq.) is buffer solution

$$\text{pH} = \text{pK}_a + \log \left(\frac{2.5}{7.5} \right) = 6 + \log \left(\frac{1}{3} \right) = 5.52$$

2. **Ans (9.35)**



$$\text{m moles} \quad 10 \quad 10$$

$$\text{After reaction} \quad 10 \text{ m moles}$$

$$\text{of NaA(aq.)}$$

$$\therefore \text{pH} = \frac{1}{2} \text{pK}_w + \frac{1}{2} (\text{pK}_a + \log C)$$

$$= 7 + \frac{1}{2} \left[6 + \log \left(\frac{10}{200} \right) \right] = 9.35$$

4. **Ans (11.00)**

$$x = \text{NH}_3$$

$$y = \text{NO}$$

$$z = \text{H}_2\text{O}$$

$$A = \text{H}_2$$

$$B = \text{O}_2$$

$$C = \text{NO}_2$$

$$E = \text{HNO}_2$$

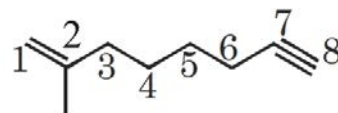
$$D = \text{HNO}_3$$

$$\text{Non bonding } e^- \left\{ \begin{array}{l} X = \text{NH}_3 = 1 \\ Y = \text{NO} = 3 \\ C = \text{NO}_2 = 5 \end{array} \right\} \Rightarrow T_{\text{total}} = 9.00$$

$$\text{Sum of M.W. of } \left\{ \begin{array}{l} z = \text{H}_2\text{O} = 18 \\ y = \text{NO} = 30 \\ D = \text{HNO}_3 = 63 \end{array} \right\} \Rightarrow 111 = M$$

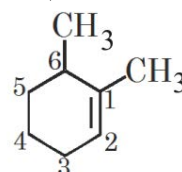
$$\text{So, } \frac{M-1}{10} = \frac{111-1}{10} = 11.00$$

5. **Ans (0.25)**



2-Methyloct-1-en-7-yne

6. **Ans (0.33)**

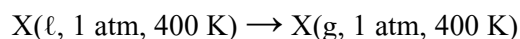
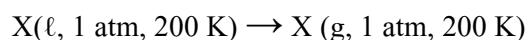


1,6-Dimethylcyclohex-1-ene

PART-2 : CHEMISTRY

SECTION-II (ii)

7. **Ans (56)**



$$\Delta_r G = \Delta H - 200 \times \Delta S$$

$$\Delta S = 1 \times \int_{200}^{400} 180 \frac{dT}{T} + \frac{100}{400} \times 1000 + \int_{400}^{200} 80 \frac{dT}{T}$$

$$\Delta S = 250 + 100 \ln 2 = 320$$

$$\Delta H = 1 \times \int_{200}^{400} 180 dT + 100 \times 1000 + 1 \times \int_{400}^{200} 80 dT$$

$$= 1000 \times 100 + 100 \times 200 = 1,20,000 \text{ J}$$

$$\Delta H = 120 \text{ kJ}$$

$$\Delta_r G = 120 - 200 \times 320 \times 10^{-3} = 56 \text{ kJ}$$

8. **Ans (1)**



9. **Ans (5)**

$-\text{NO}$, $-\text{B}(\text{OH})_2$, SO_2Me , $-\text{CN}$, $-\text{CH}_2\text{Cl}$ are deactivating groups.

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (B)**

Let $\text{trace}(A) = a$ and $\text{trace}(B) = b$. $|A| = 0$ or else we have $AB - BA = I$, which is not possible.

We have,

$$\begin{aligned}
 &(\det(A + 2B) - \det(B + 2A))I \\
 &= (a + 2b)(A + 2B) - (A + 2B)^2 - (b + 2a)(B + 2A) \\
 &\quad + (B + 2A)^2 \\
 &= (a + 2b)(A + 2B) - (b + 2a)(B + 2A) + 3(A^2 - B^2) \\
 &= (a + 2b)(A + 2B) - (b + 2a)(B + 2A) + 3(aA - (bB - 2I)) \\
 &= (a + 2b)(A + 2B) - (b + 2a)(B + 2A) + 3aA - 3bB \\
 &\quad + 6I \\
 &= aA + 2aB + 2bA + 4bB - bB - 2bA - 2aB - 4aA + 3aA - 3bB + 6I = 6I
 \end{aligned}$$

2. **Ans (C)**

$$\begin{aligned}
 y &= (\tan^4 x + \tan^2 x - 1)/(\tan^4 x + \tan^2 x + 1) \\
 y &= 1 - 2/(\tan^4 x + \tan^2 x + 1) \\
 \Rightarrow y &\in [-1, 1]
 \end{aligned}$$

3. **Ans (A)**

Let the focus be (h, k) then equation of the tangent at vertex is $y = -\frac{k}{h}(x - h)$

$$\text{i.e. } kx + hy - hk = 0$$

$$\text{its distance from } (h, k) = \left| \frac{hk}{\sqrt{h^2 + k^2}} \right| = a$$

$$\therefore \text{ the locus is } a^2(x^2 + y^2) = x^2 y^2 \quad \text{i.e.}$$

$$\frac{1}{x^2} + \frac{1}{y^2} = \frac{1}{a^2}, \quad x > 0, y > 0$$

4. **Ans (A)**

$$\text{Write equation } 4\sin^2\left(\frac{\pi}{6} - x\right)(10 - (2\cos x + 1)^2) = 40$$

From which it is immediate to get

$$\sin^2\left(\frac{\pi}{6} - x\right) = 1 \text{ and } 2\cos x + 1 = 0$$

$$\text{And so, as already given, } x = \frac{2\pi}{3} + 2k\pi$$

PART-3 : MATHEMATICS

SECTION-I (ii)

5. **Ans (B,C)**

Using AM-GM inequality

$$\begin{aligned}
 \int_0^1 x^2 dx + \int_0^1 f^2(x^2) dx &\geq \int_0^1 2xf(x^2) dx = \int_0^1 f(t) dt \\
 \Rightarrow f^2(x^2) &= x^2 \Rightarrow f(x^2) = x \text{ or } -x
 \end{aligned}$$

Here $f(x^2) = -x$ is rejected

6. **Ans (A,D)**

$$U(a, b) = {}^6C_a \cdot {}^6C_b \cdot {}^6C_{6-a-b}$$

S is equivalent of selection of six objects from 18 different objects.

$$T(a, 3) = {}^6C_a \cdot {}^6C_3 \cdot {}^6C_{|a-3|}$$

Using the values of $0 \leq a \leq 6$, we get

$$\begin{aligned}
 T(1, 3) &= T(2, 3) = T(4, 3) = T(5, 3) \text{ and } T(0, 3) \\
 &= T(3, 3) = T(6, 3)
 \end{aligned}$$

7. **Ans (B,C,D)**

$$\text{A. } f(1) \dots f(n) = 10 \text{ or } 0.1 \Rightarrow n = 18 \text{ or } 3$$

$$\text{B. } N = 2^{a_1} \cdot 3^{a_2} \cdot 5^{a_3} \dots$$

$$a_1 = 3, a_2 = 2, a_3 = 1 \text{ then } N \text{ is smallest}$$

$$\text{C. If } [x] = 2$$

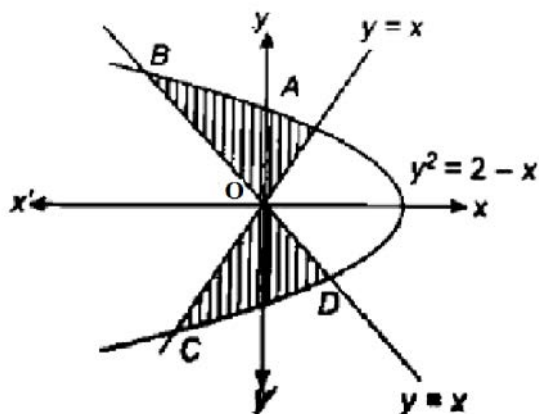
$$\log_2 3 = |x - 4| \Rightarrow x = 4 - \log_2 3$$

$$\text{If } [x] = 3 \Rightarrow \log_3 2 = |x - 4| \Rightarrow x = 4 - \log_3 2$$

$$\text{If } [x] = 4 \Rightarrow \log_4 1 = |x - 4| \Rightarrow x = 4$$

$$\begin{aligned}
 \text{D. } \frac{1}{\cos 11^\circ \cos 19^\circ} - \frac{2 \sin 19^\circ}{\cos 19^\circ} &= \frac{1 - 2 \sin 19^\circ \cos 11^\circ}{\cos 11^\circ \cos 19^\circ} \\
 &= \frac{\sin 30^\circ - \sin 8^\circ}{\cos 11^\circ \cos 19^\circ} = 2 \tan 11^\circ
 \end{aligned}$$

8. Ans (C,D)



Let $A = (1, 1)$ & $B = (-2, 2)$

maximum value of $x^2 + y^2$ corresponding to B is 8

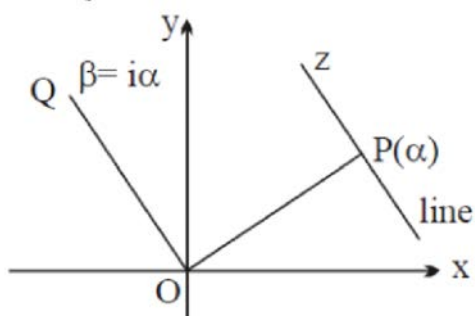
Minimum value of $x^2 + y^2$ corresponding to O is 0

integral points in R are : $(1, 1)$ $(1, -1)$ $(0, 0)$ $(0, 1)$

$(0, -1)$ $(-1, 1)$ $(-1, -1)$ $(-2, 2)$ $(-2, -2) \Rightarrow 9$ points

9. Ans (B,D)

Required line is passing through $P(\alpha)$ and parallel to the vector $\vec{OQ}$



hence $z = \alpha + i\lambda\alpha, \lambda \in \mathbb{R}$

$\frac{z - \alpha}{\alpha}$ = purely imaginary

$\Rightarrow \operatorname{Re}\left(\frac{z - \alpha}{\alpha}\right) = 0 \Rightarrow (B)$

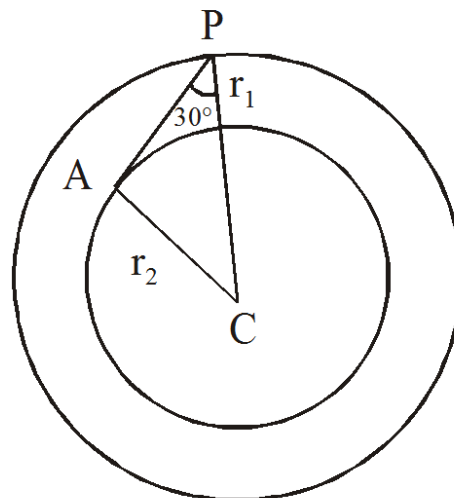
also $\frac{z - \alpha}{\alpha} + \frac{\bar{z} - \bar{\alpha}}{\bar{\alpha}} = 0$

$\bar{\alpha}(z - \alpha) + \alpha(\bar{z} - \bar{\alpha}) = 0$

$\bar{\alpha}z + \alpha\bar{z} - 2|\alpha|^2 = 0 \Rightarrow (D)$

10. Ans (C)

From triangle PAC



$$\sin 30^\circ = \frac{r_2}{r_1}$$

$$\Rightarrow r_1 = 2r_2$$

Similarly $r_2 = 2r_3$

$$r_3 = 2r_4 \dots\dots$$

$\therefore r_1, r_2, r_3, \dots\dots$ are in G.P. with common ratio $\frac{1}{2}$

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n r_i = r_1 + r_2 + r_3 + \dots$$

$$= r_1 + \frac{r_1}{2} + \frac{r_1}{4} + \frac{r_1}{8} + \dots$$

$$= 2r_1 = 40$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. Ans (2.71)

Equation of tangent at $(x_0, f(x_0)) : y - f(x_0)$

$$= f'(x_0)(x - x_0)$$

∴ intersection point with x-axis :

$$\left(x_0 - \frac{f(x_0)}{f'(x_0)}, 0 \right)$$

Equation of tangent at $(x_0, g(x_0)) :$

$$y - g(x_0) = g'(x_0)(x - x_0)$$

∴ intersection point with x-axis :

$$\left(x_0 - \frac{g(x_0)}{g'(x_0)}, 0 \right)$$

$$\therefore \frac{f(x_0)}{f'(x_0)} = \frac{g(x_0)}{g'(x_0)}$$

$$\Rightarrow \frac{f'(x_0)g(x_0) - g'(x_0)f(x_0)}{g^2(x_0)} = 0$$

$$\Rightarrow \frac{f(x_0)}{g(x_0)} = K \Rightarrow K = 2$$

$$\text{Also, } g(x) = \int_{-\infty}^x 2g(x)dx \Rightarrow g'(x) = 2g(x)$$

$$\Rightarrow g(x) = \frac{1}{2}e^{2x}$$

2. Ans (0.25)

Question Stem for Solution Nos.11 and 12

Equation of tangent at $(x_0, f(x_0)) : y - f(x_0)$

$$= f'(x_0)(x - x_0)$$

∴ intersection point with x-axis :

$$\left(x_0 - \frac{f(x_0)}{f'(x_0)}, 0 \right)$$

Equation of tangent at $(x_0, g(x_0)) :$

$$y - g(x_0) = g'(x_0)(x - x_0)$$

∴ intersection point with x-axis : $\left(x_0 - \frac{g(x_0)}{g'(x_0)}, 0 \right)$

$$\therefore \frac{f(x_0)}{f'(x_0)} = \frac{g(x_0)}{g'(x_0)}$$

$$\Rightarrow \frac{f'(x_0)g(x_0) - g'(x_0)f(x_0)}{g^2(x_0)} = 0$$

$$\Rightarrow \frac{f(x_0)}{g(x_0)} = K \Rightarrow K = 2$$

$$\text{Also, } g(x) = \int_{-\infty}^x 2g(x)dx \Rightarrow g'(x) = 2g(x)$$

$$\Rightarrow g(x) = \frac{1}{2}e^{2x}$$

3. Ans (300.00)

Let $f(1) = a, f(2) = b, f(3) = c$

Given $a + b + c = 1100$ and $S^3 = (a + 2b + 3c)^3$
 $= 162 abc$

$$\Rightarrow \frac{a + 2b + 3c}{3} = \sqrt[3]{6abc} \quad \text{Which is the case of}$$

$$AM = GM$$

when $a = 2b = 3c \Rightarrow a = 600, b = 300, c = 200$

Harmonic mean of $a, b,$

$$c = \frac{3}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \frac{3}{\frac{1}{600} + \frac{1}{300} + \frac{1}{200}} = 300$$

4. Ans (1200.00)

Arrange 600 red balls and there will be 599 gaps

(excluding initial and final). So, minimum value

will be $x_1 = 599 - 300 - 200 = 99$

For maximum arrange all 1100 and there will be

1099 gaps (excluding initial and final).

So, maximum value will be $x_2 = 1101$

So, $x_1 + x_2 = 1200$

5. Ans (0.33)

$$P_n = \frac{1}{3}P_{n-2} + \frac{2}{3}P_{n-1} \Rightarrow \frac{P_n - P_{n-1}}{P_{n-1} - P_{n-2}} = \frac{-1}{3}$$

∴ $P_2 - P_1, P_3 - P_2, P_4 - P_3, \dots$ are in G.P.

$$\text{Also, } P_2 - P_1 = \frac{1}{9} \therefore P_n - P_{n-1} = \frac{1}{9} \left(\frac{-1}{3} \right)^{n-2}$$

$$\Rightarrow P_n - P_{n-1} = \frac{1}{9} \left(\frac{-1}{3} \right)^{n-2},$$

$$P_{n-1} - P_{n-2} = \frac{1}{9} \left(\frac{-1}{3} \right)^{n-3}, \dots, P_2 - P_1 = \frac{1}{9}$$

$$\Rightarrow P_n - P_1 = \frac{1}{9} \left[\left(\frac{-1}{3} \right)^{n-2} + \left(\frac{-1}{3} \right)^{n-3} + \dots + 1 \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} P_n = P_1 + \frac{1}{12}$$

6. Ans (0.75)

Question Stem for Solution Nos. 15 and 16

$$P_n = \frac{1}{3}P_{n-2} + \frac{2}{3}P_{n-1} \Rightarrow \frac{P_n - P_{n-1}}{P_{n-1} - P_{n-2}} = \frac{-1}{3}$$

$\therefore P_2 - P_1, P_3 - P_2, P_4 - P_3, \dots$ are in G.P.

$$\text{Also, } P_2 - P_1 = \frac{1}{9} \therefore P_n - P_{n-1} = \frac{1}{9} \left(\frac{-1}{3} \right)^{n-2}$$

$$\Rightarrow P_n - P_{n-1} = \frac{1}{9} \left(\frac{-1}{3} \right)^{n-2},$$

$$P_{n-1} - P_{n-2} = \frac{1}{9} \left(\frac{-1}{3} \right)^{n-3}, \dots, P_2 - P_1 = \frac{1}{9}$$

$$\Rightarrow P_n - P_1 = \frac{1}{9} \left[\left(\frac{-1}{3} \right)^{n-2} + \left(\frac{-1}{3} \right)^{n-3} + \dots + 1 \right]$$

$$\Rightarrow \lim_{n \rightarrow \infty} P_n = P_1 + \frac{1}{12}$$

PART-3 : MATHEMATICS

SECTION-II (ii)

7. Ans (8)

$$\sin x + \cos x = t,$$

$$\tan x + \cot x = 1/\sin x \cdot \cos x = 2/(t^2 - 1)$$

$$\sec x + \operatorname{cosec} x = (\sin x + \cos x) / \sin x \cdot \cos x \\ = 2t / (t^2 - 1)$$

$$f(t) = |t + (2/(t^2 - 1)) + (2t/(t^2 - 1))|$$

$$= |t + (2/(t - 1))|$$

$$\text{min. at } 1 - \sqrt{2} \Rightarrow f(t) = 2\sqrt{2} - 1 \Rightarrow k = 8$$

8. Ans (511)

$$y = (1/(x+1)) - (1/(x+2))$$

$$\Rightarrow y_n = (-1)^n \cdot n! [(1/(x+1)^{n+1}) - (1/(x+2)^{n+1})]$$

$$\text{Put } x = 0 \text{ and } n = 8$$

$$\Rightarrow y_8 = 8! [2^9 - 1]/2^9$$

9. Ans (4)

Let position vector of ABCDEFGH be origin,

$$\vec{a}, \vec{b}, \vec{a} + \vec{b}, \vec{c}, \vec{a} + \vec{c}, \vec{b} + \vec{c}, \vec{a} + \vec{b} + \vec{c}$$

respectively

position vector of M is $\frac{\vec{a} + \vec{b}}{2}$ and of M' will

$$\text{be } \vec{c} + \frac{\vec{a} + \vec{b}}{2} \text{ and let } O = \frac{\vec{a} + \vec{b}}{2} + \mu \vec{c}$$

$$\mu = \frac{1}{5} \Rightarrow 4 \vec{MO} = \vec{OM'}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Advanced)

FULL SYLLABUS

20-05-2023

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	B,D	A,C	A,B,D	A,B,C	B	B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	B	D	D		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	163.00 to 165.00	1.85 to 1.95	5.00	20.00	2.88	2.86
SECTION-II (ii)	Q.	7	8	9			
	A.	40	4	35			

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	A,B,C,D	B,C,D	C,D	A,B,C	A,B	A,B,C,D
SECTION-I (ii)	Q.	7	8	9	10		
	A.	B	A	C	A		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	8.01	5.10	5.90	1.60	5.50	3.25
SECTION-II (ii)	Q.	7	8	9			
	A.	25	23	4			

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C,D	B,C,D	A,B,C,D	A,B,D	B,C,D	A,B,C
SECTION-I (ii)	Q.	7	8	9	10		
	A.	A	B	A	B		
SECTION-II (i)	Q.	1	2	3	4	5	6
	A.	1.00	0.00	4.50	27.50	0.00	16.00
SECTION-II (ii)	Q.	7	8	9			
	A.	3	3	100			

HINT – SHEET

PART-1 : PHYSICS

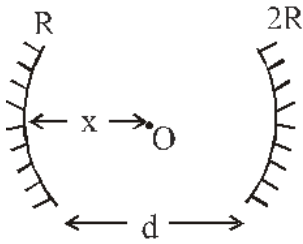
SECTION-I (i)

2. **Ans (A,C)**
Theoretical

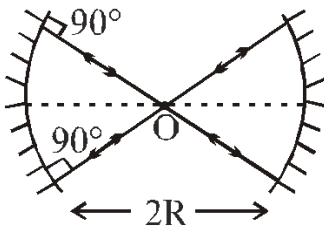
3. **Ans (A,B,D)**

The magnet is repelled by the eddy currents induced in the copper block. Faster the magnet moves greater is the retardation of the magnet. Due to this the magnet starts moving gains a maximum speed and reaches the magnet with negligible speed.

4. Ans (A,B,C)



(a) for $d = 2R \Rightarrow x = R \Rightarrow O$ will be centre of curvature. Hence image will coincide at O .

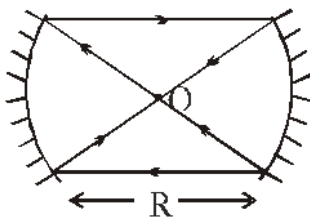


(b) for $d = R$ and $x = \frac{R}{2}$ using mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{2}{-R} = \frac{2}{-R}$$

$v = \infty$ and external ray diagram is shown

as below

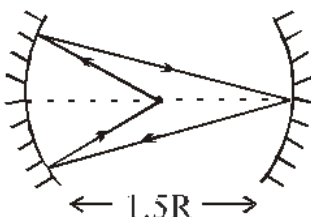


(c) for $d = 1.5R$ & $8x = \frac{3R}{4}$ using mirror

$$\text{formula } \frac{1}{v} + \frac{1}{u} = \frac{1}{f} \Rightarrow \frac{1}{v} + \frac{4}{(-3R)} = \frac{2}{-R}$$

$v = -1.5R$ and external ray diagram is

shown as below



5. Ans (B)

$$i = i_0 + i_1 \sin \omega t$$

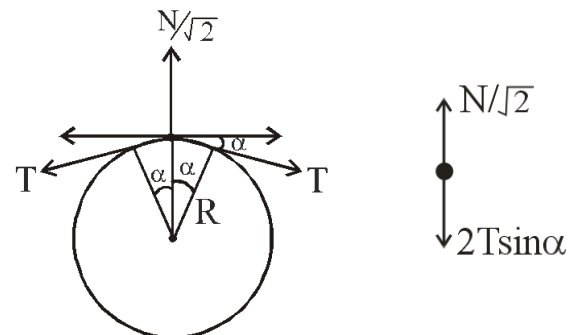
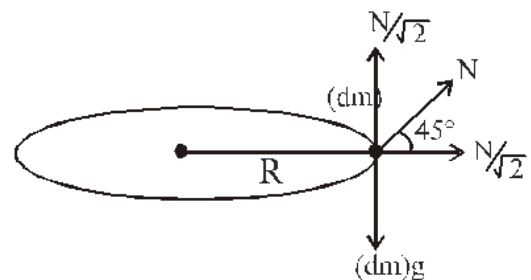
$$i_{\text{rms}} = \sqrt{\frac{\int (i_0 + i_1 \sin \omega t)^2 dt}{\int dt}}$$

$$= \sqrt{\frac{\int i_0^2 dt + \int i_1^2 \sin^2 \omega t dt + 2i_0 i_1 \int \sin \omega t dt}{\int dt}}$$

$$= \sqrt{i_0^2 + \frac{i_1^2}{2} + 2i_0 i_1 (0)} = \sqrt{\frac{2i_0^2 + i_1^2}{2}}$$

$$\langle i_{\text{avg}} \rangle = \frac{\int (i_0 + i_1 \sin \omega t) dt}{\int dt} = i_0 + \frac{i_1 \int \sin \omega t dt}{\int dt} = i_0$$

6. Ans (B,C,D)



$$\frac{M}{\sqrt{2}} = (dm)g$$

$$2T \sin \alpha = \frac{N}{\sqrt{2}} = (dm)g$$

$$2T (\sin \alpha) = (\lambda) (R2\alpha) g$$

$$\sin \alpha \approx \alpha$$

$$T = \lambda R g = \frac{Mg}{2\pi}$$

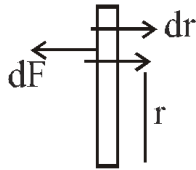
$$\text{Also } U = \frac{1}{2} \frac{(\text{stress})^2}{Y} = \text{volume}$$

$$= \frac{1}{2} \left(\frac{Mg}{2\pi A} \right)^2 \frac{1}{Y} A (2\pi R) = \frac{M^2 g^2 R}{2\pi Y A}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. Ans (B)



$$\tau \int d\tau = \int r dF$$

$$\int_0^R r (2S dr) = 2S \int_0^R r dr = SR^2$$

$$\tau = SR^2 = I\alpha = \frac{MR^2\alpha}{3}$$

$$\Rightarrow \alpha = \frac{3S}{M}$$

$$\text{Also } \omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\omega = \sqrt{2 \left(\frac{3S}{M} \right) 2\pi} = \sqrt{\frac{12S\pi}{M}}$$

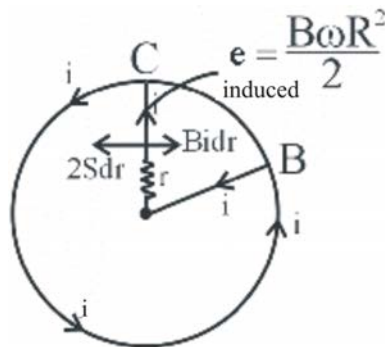
OR

using energy $\Delta E = \Delta K$

$$(2S)\pi R^2 = \frac{1}{2} I \omega^2$$

$$\omega = \sqrt{\frac{12S\pi}{M}}$$

8. Ans (B)



At equilibrium

$$2Sdr = Bidr$$

$$2S = B \left(\frac{B\omega R^2}{2R} \right)$$

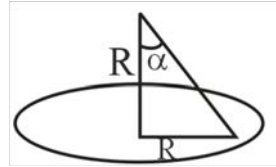
$$\omega = \frac{4Sr}{B^2 R^2}$$

9. Ans (D)

$$F_{\text{lux}} = \frac{q}{2\epsilon_0} - \frac{q}{2\epsilon_0} (1 - \cos \alpha)$$

$$F_{\text{lux}} = \frac{q}{2\epsilon_0} (1 - 1 + \cos \alpha)$$

$$F_{\text{lux}} = \frac{q}{2\sqrt{2}\epsilon_0}$$



$$\cos \alpha = \frac{1}{\sqrt{2}}$$

$$\therefore \alpha = 45^\circ$$

10. Ans (D)

$$E = \frac{\sigma}{2\epsilon_0} - \frac{\sigma}{2\epsilon_0} (1 - \cos \alpha)$$

$$= \frac{\sigma}{2\epsilon_0} (1 - 1 + \cos \alpha) = \frac{\sigma}{2\sqrt{2}\epsilon_0}$$

$$F = ma$$

$$qE = ma$$

$$\frac{q\sigma}{2\sqrt{2}\epsilon_0} = ma$$

$$\Rightarrow a = \frac{2\sigma}{2\sqrt{2}\epsilon_0 m}$$

PART-1 : PHYSICS

SECTION-II (i)

1. Ans (163.00 to 165.00)

$$48.36 \times 1.6 \times 10^{-19} \times \frac{5}{32} = \frac{hc}{\lambda} = \frac{6.6 \times 10^{-34} \times 3 \times 10^8}{\lambda}$$

2. Ans (1.85 to 1.95)

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}}$$

$$\text{Here } hv = 48.36 \text{ eV}$$

$$KE = hv - \phi_0 = 48.36 - 6 \text{ eV} = 42.36 \text{ eV}$$

$$K = 42.36 \times 1.6 \times 10^{-19} \text{ J}$$

using values

$$\lambda = 1.879 \times 10^{-10} \text{ m} = 1.879 \text{ \AA}$$

5. Ans (2.88)

Reading = M.SD + (L.C) × (coinciding division)

Hence

$$10 \text{ V.SD} = 9 \text{ M.SD and } 1 \text{ M.SD} = 0.1 \text{ cm}$$

$$1 \text{ V.SD} = 0.9 \text{ M.SD} = 0.09 \text{ cm}$$

$$L - C = 1 \text{ M.SD} - 1 \text{ V.SD} = 0.01 \text{ cm}$$

$$\text{Reading} = (2.8) + (0.01)(8) = 2.88$$

6. Ans (2.86)

$$10 \text{ V.SD} = 11 \text{ M.SD}$$

$$1 \text{ V.SD} = 1.1 \text{ M.SD} \text{ \& } 1 \text{ MSD} = 0.1 \text{ cm} = 0.11 \text{ cm}$$

$$\text{Reading} = (2.90) + (-0.01) \times 4 = 2.86$$

PART-1 : PHYSICS

SECTION-II (ii)

7. Ans (40)

$$I = I_{\max} \cos^2 \left(\frac{\Delta\phi}{2} \right)$$

$$I = \frac{3}{4} I_{\max}$$

$$\Rightarrow \cos^2 \left(\frac{\Delta\phi}{2} \right) = \frac{3}{4}$$

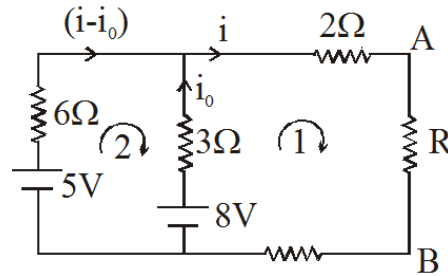
$$\Rightarrow \frac{\Delta\phi}{2} = 30^\circ, \frac{150^\circ}{\pi}$$

we will select 150° or for 2nd time (ATQ)

$$\frac{\Delta\phi}{2} = \frac{5\pi}{6} \Rightarrow \Delta\phi = \frac{5\pi}{3} \text{ or } \frac{40\pi}{24}$$

Hence $n = 40$

8. Ans (4)



using KVL in loop 1 $8 - 3i_0 - i(2 + R) = 0$

.....(1)

KVL in loop 2

$$5 - 6(i - i_0) - 8 + 3i_0 = 0 \quad \text{.....(2)}$$

solving eq (1) and (2) we get

$$i = \frac{7}{4 + R}$$

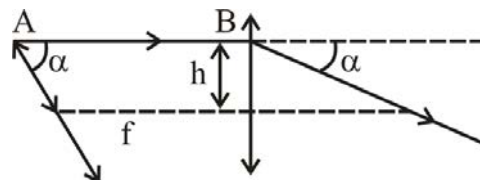
$$\text{Now P (across R)} = i^2 R = \left(\frac{7}{4 + R} \right)^2 R$$

$$P = \frac{49R}{(4 + R)^2}$$

for P-max $(R - 4) = 0$

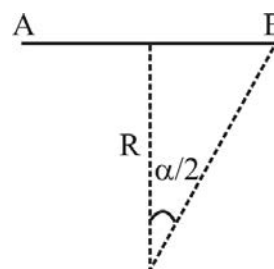
$$R = 4$$

9. Ans (35)



$$f = \frac{h}{\tan \alpha} = \alpha = \frac{2\pi}{20} \Rightarrow h = f\alpha$$

$$AB = f + h \tan \alpha = f + h\alpha = f + f\alpha^2 = f(1 + \alpha^2)$$



$$\frac{AB/2}{R} = \tan \frac{\alpha}{2}$$

$$AB = 2R \tan \frac{\alpha}{2} = R\alpha$$

$$R = \frac{f(1 + \alpha^2)}{\alpha} = \frac{10 \left(1 + \left(\frac{2\pi}{200} \right)^2 \right)}{\frac{2\pi}{20}}$$

$$= 10 \left(1 + \frac{10}{100} \right) \times \frac{10}{\pi} = 10 \times \frac{11}{10} \times \frac{10}{22} = 35$$

$$R = 35 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,B,C,D)

$$\text{Molar conductivity} = 100 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\lambda(\text{ClCH}_2\text{COO}^-) = 50 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$\lambda(\text{H}^+) = 350 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$$

$$C = 0.012 \text{ mol dm}^{-3}$$

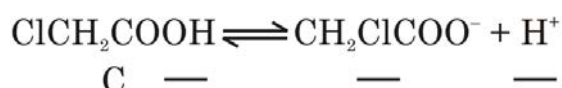
For the dissociation of ClCH_2COOH

$$\lambda_{(\text{ClCH}_2\text{COOH})}^\infty = \lambda_{(\text{ClCH}_2\text{COO}^-)}^\infty + \lambda_{(\text{H}^+)}^\infty = 400 \text{ cm}^2 \text{ mol}^{-1}$$

$$\alpha = \frac{\lambda^\circ}{\lambda^\infty}$$

$$\alpha = \frac{100}{400} = 0.25$$

Now,



$$K_a = \frac{C(1-\alpha)(\alpha)^2}{C(1-\alpha)}$$

$$K_a = \frac{0.012(0.25)^2}{(1-0.25)} = 10^{-3}$$

$$\text{pK}_a = -\log 10^{-3} = 3$$

$$[\text{H}]^+ = C\alpha$$

$$[\text{H}]^+ = 0.012 \times 0.25 = 3 \times 10^{-3} \text{ mol dm}^{-3}$$

$$\text{pH} = 3 - \log 3$$

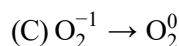
$$= 3 - 0.48$$

$$= 2.52$$

4. Ans (A,B,C)

(A) $[\text{FeCl}(\text{CN})_4(\text{O}_2)]^{-4}$ if Fe in +2, O_2^{-1} is present which is paramagnetic, and if Fe is in +3, then complex is para irrespective of ligand

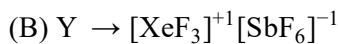
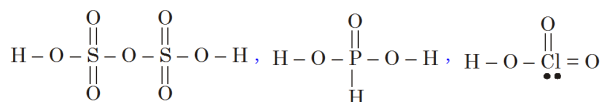
(B) In $\text{Ni}(\text{CO})_4$, pairing takes place i.e., number of unpaired electrons have reduced compared to those in Ni, so low spin



(D) $\text{X} \rightarrow \text{K}_3[\text{Cu}(\text{CN})_4]$; Cu is sp^3 hybridised

5. Ans (A,B)

(A)



(C) No axial position is present in octahedral geometry

(D) X is FeCl_2 as liberation of hydrogen prevents formation of FeCl_3

6. Ans (A,B,C,D)



$$\text{A}^{+2} \text{ B}^- \quad \text{B}^- \sim \text{F}^- \quad x+y = \left[\frac{(\text{edge length})\sqrt{3}}{4} \right]$$

A^{+2} occupy corners and face centres

B^- All tetrahedral voids

Max. distance between TVs

$$\begin{aligned} (\text{two B}^-) &= \frac{(\text{edge length})}{2}(\sqrt{3}) \\ &= \frac{4}{\sqrt{3}}(\text{X} + \text{Y}) \frac{\sqrt{3}}{2} \\ &= 2(\text{X} + \text{Y}) \end{aligned}$$

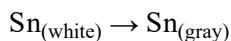
Min distance (two B^-)

$$= \frac{\text{Edge length}}{2} = \frac{2}{\sqrt{3}}(\text{X} + \text{Y})$$

PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (C)



$$\Delta H = -2.1 - 0 = -2.1 \text{ kJ}$$

$$\Delta S = 44 - 52 = -8 \text{ J/K}$$

$$\Delta G = -2.1 + \frac{8 \times 300}{1000} = -2.1 + 2.4 = 0.3 \text{ kJ}$$

$\therefore \text{Sn}_{(\text{gray})}$ is thermodynamically less stable.

$\therefore G$ of $\text{Sn}_{(\text{white})}$ will be less.

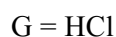
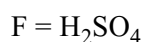
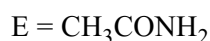
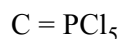
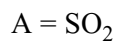
10. Ans (A)

Also on decreasing temperature, $\text{Sn}_{(\text{white})}$ converts to $\text{Sn}_{(\text{gray})}$ which makes it more brittle and expansion will be observed.

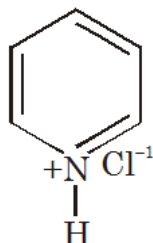
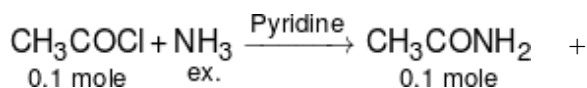
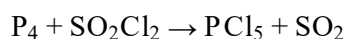
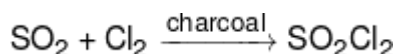
PART-2 : CHEMISTRY

SECTION-II (i)

4. Ans (1.60)



Reactions :

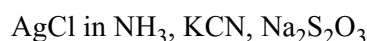


⇒ moles of NaOH

$$\text{required} = 0.2 + 0.2 = 0.4 \Rightarrow \text{weight} = 0.4 \times 40$$

$$= 16 \text{ g} = y$$

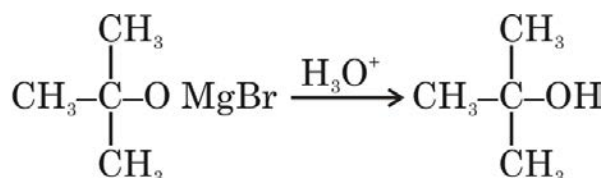
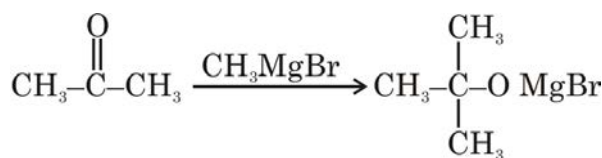
6. Ans (3.25)



PART-2 : CHEMISTRY

SECTION-II (ii)

7. Ans (25)



1 mole ethylacetate needs 2 moles MeMgBr

$$\text{moles of ethylacetate} = 10 \times 0.25 \text{ m mol}$$

$$= 2.5 \text{ m mol}$$

$$\therefore \text{moles of MeMgBr required} = 5 \text{ m mol}$$

$$\therefore 0.2 = \frac{5 \text{ m mol}}{V \text{ (in mL)}}$$

$$V = 25 \text{ mL}$$

9. Ans (4)

(iii), (iv), (v) and (vii)

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C,D)

(A) $(\text{ABA}^T)^T = \text{AB}^T\text{A}^T$ which is not symmetric

(B) $(\text{AB} - \text{BA})^T = (\text{AB})^T - (\text{BA})^T = \text{B}^T\text{A}^T - \text{A}^T\text{B}^T$

$$\therefore \text{AB} - \text{BA} \text{ is not symmetric}$$

(C) $B = |A| \frac{\text{adj. } A}{|A|} = \text{adj. } A$

$$\text{adj } A^T - B = \text{adj } A^T - \text{adj. } (A)$$

$$(\therefore \text{adj } A^T = (\text{adj } A)^T)$$

$$((\text{adj } A)^T - \text{adj } A)^T = \text{adj. } A - (\text{adj. } A)^T$$

$$\therefore \text{adj. } (A^T) - B \text{ is skew symmetric}$$

(D) $B + A^T = O$ and $A^T = -A$

$$\Rightarrow B = A$$

$$B^{15} = A^{15}, A \text{ is skew symmetric}$$

$$\therefore A^{15} \text{ is also skew symmetric}$$

2. Ans (B,C,D)

$$\frac{x^2}{16} - \frac{y^2}{4} = 1$$

$\left(5, \frac{3}{2}\right)$ lies on the curve.

tangent at A

$$\frac{5x}{16} - \frac{3}{2} \left(\frac{y}{4}\right) = 1 \Rightarrow 5x - 6y = 16$$

tangent at B

$$5x - 6y = -6$$

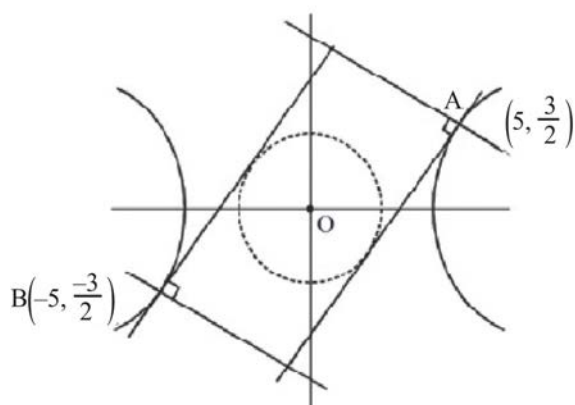
Normal at A

$$6x + 5y = \lambda$$

$$A \left(5, \frac{3}{2}\right)$$

$$30 + \frac{15}{2} = \lambda \Rightarrow \lambda = \frac{75}{2}$$

$$6x + 5y = \frac{75}{2}$$



parallelly normal at B is $6x + 5y = \frac{-75}{2}$

distance between tangents is $2r = \frac{32}{\sqrt{25+36}}$

$$\therefore r = \frac{16}{\sqrt{61}}$$

3. Ans (A,B,C,D)

Given that,

$$a_1 + a_2 \cos 2x + a_3 \sin^2 x = 0 \forall x \in \mathbb{R}$$

$$\Rightarrow a_1 + a_2(1 - 2\sin^2 x) + a_3 \sin^2 x = 0 \forall x \in \mathbb{R}$$

$$\Rightarrow (a_1 + a_2) + \sin^2 x(a_3 - 2a_2) = 0 \forall x \in \mathbb{R}$$

$$\text{So, } a_1 + a_2 = 0$$

and $a_3 = 2a_2$ must hold simultaneously

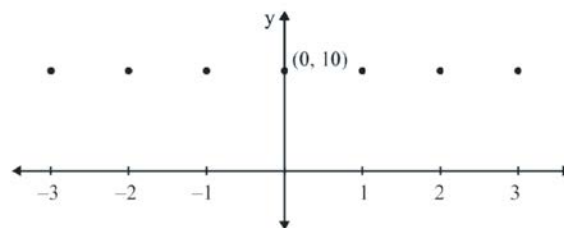
$$\text{So, } \frac{a_1}{-1} = \frac{a_2}{1} = \frac{a_3}{2} = \lambda (\neq 0)$$

$$\Rightarrow (a_1, a_2, a_3) = (-\lambda, \lambda, 2\lambda) \forall x \in \mathbb{R}_0$$

4. Ans (A,B,D)

$$f: I \rightarrow I$$

$$f(x) = 2 \sin 2\pi x - 10 \tan 5\pi x + 7 \cos 4\pi x + 3$$



$$\text{Clearly } f(x) = 10, \forall x \in I$$

$$\therefore R_f \in \{10\}$$

Graph of $f(x)$ is as below :

$f(x)$ is periodic

$$f(f(f(x))) = 10$$

$$f(f(x)) = 10$$

5. Ans (B,C,D)

Let

$$J = \int_0^{1/2} \left(\frac{1}{4} - x^2\right)^4 dx = \int_0^{1/2} \left(\frac{1}{4} - \left(\frac{1}{2} - x\right)^2\right)^4 dx$$

(Using King property)

$$\therefore J = \int_0^{1/2} (x - x^2)^4 dx = \int_0^{1/2} x^4 (1 - x)^4 dx \quad \dots (i)$$

$$\text{Put } x = 1 - y$$

Also,

$$J = - \int_1^{1/2} (1 - y)^4 y^4 dy \Rightarrow J = \int_{1/2}^1 x^4 (1 - x)^4 dx \quad \dots (ii)$$

$$\therefore (i) + (ii)$$

$$\Rightarrow 2J = \int_0^1 x^4 (1 - x)^4 dx \Rightarrow J = \frac{1}{2} \int_0^1 x^4 (1 - x)^4 dx$$

(Using Queen property)

$$\text{Put } x = \sin^2 \theta,$$

$$J = \int_0^{\pi/2} \sin^8 \theta \cdot \cos^8 \theta d\theta = \frac{(8 \times 6 \times 4 \times 2)(8 \times 6 \times 4 \times 2)}{(18 \times 16 \times 14 \times 12 \times 10 \times 8 \times 6 \times 4 \times 2)}$$

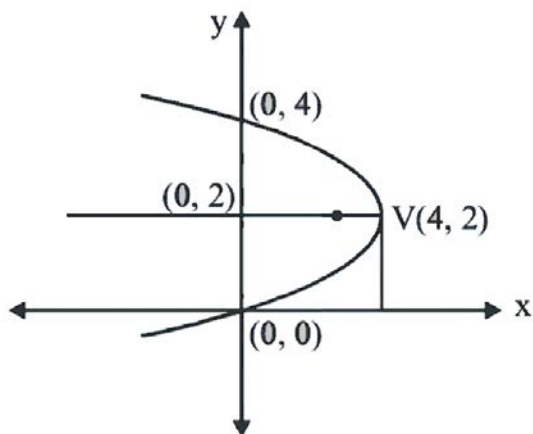
$$= \frac{(8 \times 6 \times 4 \times 2)}{(18 \times 16 \times 14 \times 12 \times 10)} = \frac{1}{1260}$$

6. Ans (A,B,C)

$$\frac{dx}{dy} = \frac{x}{y} - y$$

$$\frac{dx}{dy} - \frac{1}{y}x = -y$$

$$\text{IF} = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$



$$x \cdot \frac{1}{y} = -y + c$$

$$x = -y^2 + cy \quad 4 = -4 + 2c$$

$$x = -y^2 + 4y \quad c = 4$$

$$(x - 4) = -(y - 2)^2 \Rightarrow (y - 2)^2 = -(x - 4)$$

PART-3 : MATHEMATICS

SECTION-I (ii)

8. Ans (B)

$$\frac{dy}{dx} = -4x(y - 2)$$

$$\frac{dy}{y - 2} = -4x dx$$

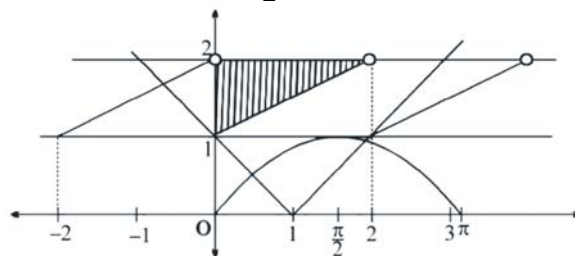
$$\ln|y - 2| = -2x^2 + C$$

$$|y - 2| = e^{-2x^2 + C} = \lambda e^{-2x^2}$$

$$\text{at } x = 0, y = 2 \Rightarrow \lambda = 0$$

$$\therefore y - 2 = 0 \Rightarrow y = f(x) = 2$$

$$7. g(x) = \max. \left\{ 1 + \left\{ \frac{x}{2} \right\}, \sin x, |x - 1| \right\}$$

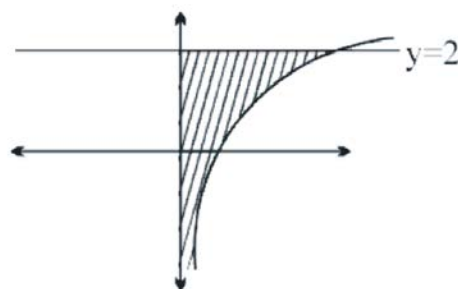


$$\text{Required area} = \frac{1}{2} \times 1 \times 2 = 1 \text{ sq. units.}$$

$$8. h(x) = \ln x$$

$$\text{Required area} = \int_{-\infty}^2 e^y dy$$

$$= (e^y)_{-\infty}^2 = e^2$$



9. Ans (A)

$$f(x) = f(2 - x)$$

$$\Rightarrow b = 2$$

$$f(x) = -x^2 + 2x + c$$

$$-x^2 + 2x + c = 0$$

$$x = 1 \pm \sqrt{1 + c}$$

$$c \in (0, 1)$$

$$2 < 1 + \sqrt{1 + c} < 1 + \sqrt{2}$$

$$\max([\alpha], [\beta]) = 2$$

10. **Ans (B)**

$$c \in (-1, 0)$$

$$x = 1 \pm \sqrt{1+c}$$

$$\min([\alpha], [\beta]) = [1 - \sqrt{1+c}] = 0$$

$$\max([\alpha], [\beta]) = [1 + \sqrt{1+c}] = 1$$

$$[\alpha] + [\beta] = 1 < 2$$

$$f(x)_{\max} = f(1) = 1 + c$$

$$0 < f(1) < 1$$

PART-3 : MATHEMATICS

SECTION-II (i)

1. **Ans (1.00)**

$$y^2 = 8x$$

$$y^3 = 8xy \Rightarrow y^3 = -8 \Rightarrow y = -2$$

$$\{\because xy = -1\}$$

$$\Rightarrow x = \frac{1}{2}$$

$$y^2 = 8x \Rightarrow 2yy' = 8 \Rightarrow y'|_{(1/2, -2)} = -2$$

$$\therefore \text{tangent at } \left(\frac{1}{2}, -2\right) \text{ is, } y + 2 = -2 \left(x - \frac{1}{2}\right)$$

$$2x + y + 1 = 0$$

$$\therefore y \text{ intercept} = -1$$

2. **Ans (0.00)**

$$\therefore y' = \cos(x+y)(1+y')$$

$$-2 = \cos(x+y)(-1)$$

$$\cos(x+y) = 2$$

Not possible

$\therefore$ no such tangent is possible

4. **Ans (27.50)**

$$S_1 = \sum_{k=1}^{10} k(k-1) \frac{10!}{k(k-1)(k-2)!(10-k)!}$$

$$= 90 \sum_{k=2}^{10} \frac{8!}{(k-2)!(8-(k-2))!} = 90 \sum_{k=2}^{10} {}^8C_{k-2} = 90 \times 2^8$$

$$\text{Now, } S_2 = \sum_{k=1}^{10} \frac{k(10)!}{k(k-1)!(9-(k-1))!} = 10 \sum_{k=1}^{10} {}^9C_{k-1} = 10 \times 2^9$$

$$\text{and } S_3 = \sum_{k=1}^{10} (k(k-1) + k) \frac{10!}{k! (10-k)!}$$

$$= \left(\sum_{k=1}^{10} k(k-1) \cdot {}^{10}C_k \right) + \left(\sum_{k=1}^{10} k \cdot {}^{10}C_k \right)$$

$$= S_1 + S_2 = (90 \times 2^8) + (10 \times 2^9)$$

$$= (90 \times 2^8) + (20 \times 2^8) = 110 \times 2^8 = 55 \times 2^9$$

5. **Ans (0.00)**

$$I = \int_1^{-1} g(1+x)dx \quad \text{Put } 1+x = 2-y$$

$$dx = -dy$$

$$I = - \int_0^2 g(2-y) dy \quad \text{and } y = 1-x$$

$$I = - \int_0^2 g(y)dy \quad \because \int_0^a f(x)dx = \int_0^a f(a-x)dx$$

$$\text{Add above equation, we get } I = 0$$

6. **Ans (16.00)**

$$2I = - \int_0^2 f(x^4)dx = 2 \int_0^2 f(x^4)dx = 2 \int_0^2 (3x^2 - g(2-x))dx$$

$$= 2 \int_0^2 3x^2dx - 2 \int_0^2 g(2-x)dx = 2 \cdot \left[\frac{3x^3}{3} \right]_0^2 - 0 = 16$$

PART-3 : MATHEMATICS
SECTION-II (ii)
7. Ans (3)

 Let $f(x) = \tan^{-1}x, x \in [a, b]$
 $\therefore$ Using LMVT, we get $\frac{\tan^{-1}b - \tan^{-1}a}{b-a} = \frac{1}{1+c^2}$,

 where $0 < a < c < b < \sqrt{3}$

 So, $1 < \left(\frac{b-a}{\tan^{-1}b - \tan^{-1}a} \right) < 4$

 As, $\frac{1}{4} < \frac{1}{1+c^2} < 1$
 $\Rightarrow$ The greatest possible integral value is 3.

8. Ans (3)

$$\int_0^{\pi/2} \frac{dx}{\cos^2 x (1 + \sqrt{\tan x})^4} = \int_0^{\pi/2} \frac{\sec^2 x}{(1 + \sqrt{\tan x})^4} dx$$

 Put $\tan x = t^2 \Rightarrow \sec^2 x dx = 2t dt$

$$\int_0^\infty \frac{2t}{(1+t)^4} dt = 2 \int_0^\infty \left(\frac{1}{(1+t)^3} - \frac{1}{(1+t)^4} \right)$$

$$dt = 2 \left(\frac{-1}{2(1+t)^2} + \frac{1}{3(1+t)^3} \right)_0^\infty$$

$$= \left(\frac{-1}{2(1+t)^2} + \frac{1}{3(1+t)^3} \right)_0^\infty$$

$$= 2 \left(0 - \left(\frac{-1}{2} + \frac{1}{3} \right) \right) = 2 \left(\frac{1}{2} - \frac{1}{3} \right) = \frac{1}{3}$$

9. Ans (100)

$$[\{x\}] = 0$$

$$f(x) = (3 - x^7)^{\frac{1}{7}}$$

$$f(f(x)) = \left(\left(3 - (3 - x^7)^{\frac{1}{7}} \right)^7 \right)^{\frac{1}{7}} = x$$

$$f(f(x)) = x \Rightarrow f^{-1}(x) = f(x)$$

$$\therefore f^{-1}(50) = f(50) \text{ and } f(f(100)) = 100$$

$$\therefore f^{-1}(50) - f(50) + f(f(100)) = 100$$